Sourcing recommendation: CNC aluminium enclosure for final assembly near Chennai

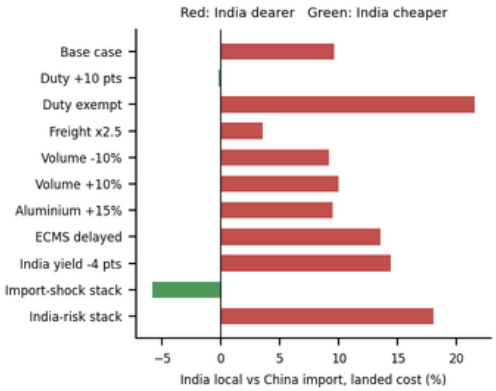
Recommendation

Qualify anIndiansupplier as a second source now, but do not move the bulk of volume on cost alone yet. At current duty, freight and yield assumptions, a China import lands at **$3.68** per unit versus **$4.03** from a local Indian supplier (**+9.6%**); Vietnam lands at $3.97. The local option becomes cheapest only when imports are hit: import duty at or above about **20%** (modelled today: 10%), freight above about **3.3x**, or the combined import-shock case (India 5.8% cheaper).

What drives the gap

- **Yield and machining, not logistics.** India carries a $0.87/unit penalty on yield loss ($0.48 vs $0.18) and machining ($2.10 vs $1.53). China's entire import burden (freight, duty, handling, carrying cost) is $0.54/unit.
- **The ECMS incentive helps but does not close it alone.** At 50% pass-through it is worth $0.15/unit. At 86% yield, even full pass-through leaves India $0.21/unit dearer; parity needs about 95% yield at 50% pass-through, or about 92% with full pass-through (India ahead by $0.02).
- **Local supply is the stable hedge.** India's landed cost spans $4.00 to $4.40 across all 11 scenarios (10% spread) versus $3.32 to $4.32 for China (30% spread). Volume swings of 10% move cost by only about 1%.

Scenario (USD per unit)	India	China	Vietnam	Cheapest
Base case	$4.03	$3.68	$3.97	China
Duty +10 pts	$4.03	$4.04	$4.38	India
Freight x2.5	$4.04	$3.91	$4.22	China
India yield -4 pts	$4.21	$3.68	$3.97	China
Import-shock stack	$4.07	$4.32	$4.69	India
India-risk stack	$4.40	$3.73	$4.01	China



Incoterms choice for imported supply

For a China import the buyer-managed FOB cost is $3.68; CIF adds 0.3% and DDP adds 1.0% ($0.04/unit). DDP is inexpensive protection if the team wants the seller to carry clearance work and duty-change handling, but suppliers typically re-price when the duty rate moves, so duty exposure is reduced rather than removed.

Recommended next steps

- **Replaceassumptions with quotes:** RFQ two or three Indian CNC suppliers for cycle time, first-pass yield and tooling, and re-run the model.
- **Stage volume against yield milestones:** release additional share only as measured first-pass yield approaches the break-even range above.
- **Lock the incentive terms:** negotiate the share of ECMS benefit passed to the buyer in the supply agreement.
- **Verify tariff inputs:** confirm the current Indian duty on the enclosure's HS code and any preferential rate for Vietnam-origin parts.

Context not modelled:US dutieson finished devices differby assembly country (Section301 forced-laborduty from 24 Jul 2026: India 10%,China and Vietnam

12.5%, plus China's older list duties), which adds a strategic reason to localise. Inputs: LME aluminium USD 3,134/t (6 Oct 2026) and the ECMS enclosure incentive range of 6-8% are sourced; cycle times, rates, yields, tooling and freight are illustrative assumptions, not Apple data or supplier quotes. See the Sources sheet.